

Mathematical Preliminaries

P.1 Methods of Proof

P.1.1 Direct proof, contraposition, contradiction

Definition P.1 (Statements and implication). A **statement** is a declarative sentence that is either true or false. If P and Q are statements, the implication

$$P \implies Q$$

says that whenever P is true, Q is true. The statement P is the **hypothesis**, and Q is the **conclusion**. Its **converse** is $Q \implies P$ and its **contrapositive** is not $Q \implies$ not P .

The converse is a different implication and need not be true. The contrapositive is logically equivalent to the original implication.

Theorem P.2 (Contrapositive). The implication $P \implies Q$ is true if and only if not $Q \implies$ not P is true.

Proof. Suppose first that $P \implies Q$ is true. If Q is false but P were true, the implication would make Q true, a contradiction. Thus P is false. Conversely, suppose the contrapositive is true and that P is true. If Q were false, the contrapositive would make P false. Therefore Q is true. $\square$

Definition P.3 (Three proof methods). To prove $P \implies Q$:

- (i) a **direct proof** assumes P and derives Q ;
- (ii) a **proof by contraposition** assumes that Q is false and derives that P is false;
- (iii) a **proof by contradiction** assumes P and that Q is false, and derives two incompatible statements.

A proof of a statement Q by contradiction assumes that Q is false and derives a contradiction.

Example P.1 (Direct proof). Suppose $\mathbf{A}$ and $\mathbf{B}$ are square symmetric matrices of the same order and $\mathbf{AB} = \mathbf{BA}$. Then $\mathbf{AB}$ is symmetric. Indeed,

$$(\mathbf{AB})' = \mathbf{B}'\mathbf{A}' = \mathbf{BA} = \mathbf{AB}.$$

The displayed calculation starts from the hypothesis and ends at the defining equality for the conclusion.

Example P.2 (Contraposition). For a real matrix $\mathbf{A}$, if $\mathbf{A}'\mathbf{A} = \mathbf{O}$, then $\mathbf{A} = \mathbf{O}$. Prove the contrapositive. If $\mathbf{A} \neq \mathbf{O}$, some entry a_{ij} is nonzero. The j th diagonal entry of $\mathbf{A}'\mathbf{A}$ is

$$\sum_k a_{kj}^2 \geq a_{ij}^2 > 0,$$

so $\mathbf{A}'\mathbf{A} \neq \mathbf{O}$. Theorem P.2 now gives the original implication.

Example P.3 (Contradiction). There is no largest real number. If there were one, call it M . Then $M + 1$ would be a real number and $M + 1 > M$, contradicting the assertion that M is largest.

Theorem P.4 (Negating quantifiers). For any property $P(x)$ on a set D ,

$$\text{not}[\text{for every } x \in D, P(x)] \iff [\text{for some } x \in D, P(x) \text{ is false}],$$

and

$$\text{not}[\text{for some } x \in D, P(x)] \iff [\text{for every } x \in D, P(x) \text{ is false}].$$

Proof. If it is false that $P(x)$ holds for every x , at least one member of D fails the property; and the existence of such a member makes the universal claim false. Likewise, the assertion that no member has the property says exactly that every member fails it. $\square$

Definition P.5 (Counterexample). A **counterexample** to a universal statement is one object for which the claimed property fails. By Theorem P.4, producing a counterexample proves that the universal statement is false.

Example P.4 (Zero products). The claim that $\mathbf{AB} = \mathbf{O}$ implies $\mathbf{A} = \mathbf{O}$ or $\mathbf{B} = \mathbf{O}$ is false for

matrices. The two nonzero matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

satisfy $\mathbf{AB} = \mathbf{O}$. This pair is a counterexample to the universal claim.

P.1.2 Induction

Definition P.6 (Positive integers). A subset S of $\mathbb{R}$ is **inductive** if $1 \in S$ and $x + 1 \in S$ whenever $x \in S$. Define $\mathbb{N}$ to be the intersection of all inductive subsets of $\mathbb{R}$. The collection is nonempty because $\mathbb{R}$ is inductive. Every inductive set contains 1, so their intersection contains 1; and if x belongs to every inductive set, then $x + 1$ belongs to every one of them. Thus $\mathbb{N}$ is itself inductive and is contained in every inductive subset of $\mathbb{R}$.

Theorem P.7 (Principle of mathematical induction). Let $P(n)$ be a statement for each $n \in \mathbb{N}$. If

- (i) $P(1)$ is true; and

(ii) for every $k \in \mathbb{N}$, the truth of $P(k)$ implies the truth of $P(k + 1)$,
then $P(n)$ is true for every $n \in \mathbb{N}$.

Proof. Let $S = \{n \in \mathbb{N} : P(n) \text{ is true}\}$. Assumption (i) puts 1 in S , and assumption (ii) puts $k + 1$ in S whenever $k \in S$. Hence S is inductive. By Definition P.6, $\mathbb{N}$ is contained in every inductive set, so $\mathbb{N} \subseteq S$. The reverse inclusion is part of the definition of S ; therefore $S = \mathbb{N}$. $\square$

Theorem P.8 (Strong induction). Let $P(n)$ be a statement for each $n \in \mathbb{N}$. Suppose $P(1)$ is true and, for every $k \in \mathbb{N}$, the joint truth of $P(1), \dots, P(k)$ implies $P(k + 1)$. Then $P(n)$ is true for every $n \in \mathbb{N}$.

Proof. Let $Q(k)$ be the statement that $P(1), \dots, P(k)$ are all true. The hypothesis gives $Q(1)$. If $Q(k)$ is true, the hypothesis gives $P(k + 1)$, so $Q(k + 1)$ is true. Theorem P.7 gives $Q(k)$ for every k , and hence $P(k)$ for every k . $\square$

Example P.5 (A finite sum). For every $n \in \mathbb{N}$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

For $n = 1$ both sides equal 1. If the identity holds at $n = k$, then

$$1 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2},$$

which is the identity at $k+1$. Theorem P.7 completes the proof.

Example P.6 (Transpose of a finite product). If $\mathbf{A}_1, \dots, \mathbf{A}_n$ have conformable sizes, then

$$(\mathbf{A}_1 \mathbf{A}_2 \cdots \mathbf{A}_n)' = \mathbf{A}_n' \mathbf{A}_{n-1}' \cdots \mathbf{A}_1'.$$

At $n = 1$ the claim is immediate. Suppose it holds for $n = k$. The transpose rule for a product of two matrices gives

$$(\mathbf{A}_1 \cdots \mathbf{A}_k \mathbf{A}_{k+1})' = \mathbf{A}_{k+1}' (\mathbf{A}_1 \cdots \mathbf{A}_k)' = \mathbf{A}_{k+1}' \mathbf{A}_k' \cdots \mathbf{A}_1',$$

which is the claim for $k + 1$. Induction proves the identity for every finite product.

P.2 Complex Numbers and Polynomials

P.2.1 Complex arithmetic

To write

- Definition of $\mathbb{C}$, conjugate, modulus, polar form. Theorem: $\overline{zw} = \bar{z}\bar{w}$ and $|zw| = |z||w|$.
- Definition of the conjugate transpose $\mathbf{A}^*$, of a Hermitian matrix, and of a unitary matrix. These appear only in Schur's theorem and are otherwise not used.

P.2.2 Polynomials and their roots

To write

- Theorem (fundamental theorem of algebra), stated with a proof sketch and a pointer to where a full proof needs analysis this course does not assume.
- Corollary: a real polynomial has its non-real roots in conjugate pairs, so a real matrix of odd order has a real eigenvalue.
- Theorem (Vieta): the elementary symmetric functions of the roots as the coefficients, which is what makes $\text{tr } \mathbf{A} = \sum \lambda_i$ and $|\mathbf{A}| = \prod \lambda_i$ work.

P.3 Series and the Neumann Expansion

P.3.1 Geometric and power series

To write

- The geometric series and its radius of convergence, stated for the scalar case.
- Definition of a matrix power series and of convergence entrywise.

P.3.2 The Neumann series

To write

- Theorem: $\mathbf{I} - \mathbf{A}$ is nonsingular with $(\mathbf{I} - \mathbf{A})^{-1} = \sum_{k \geq 0} \mathbf{A}^k$ when every eigenvalue of $\mathbf{A}$ has modulus below one.
- Corollary: the perturbation bound behind the conditioning theorem of the elimination unit.
- Definition of $\exp(\mathbf{A})$ by its series, used once, in the remark on functions of a symmetric matrix.

P.4 Calculus in Several Variables

P.4.1 Differentiability

To write

- Definition of the partial derivative and of (total) differentiability, and the standard example separating them.
- Theorem: continuous partial derivatives imply differentiability. This is the hypothesis every result in the matrix calculus units quietly needs.

P.4.2 Taylor's theorem

To write

- Theorem: the first- and second-order expansions with remainder, in the form the second differential unit uses to classify stationary points.

P.4.3 Implicit functions and differentiation under the integral

To write

- Theorem (implicit function theorem), needed for the eigenvalue and eigenvector differentials.
- Theorem: differentiation under the integral sign, needed once, for the score of the normal likelihood.

P.5 Notation

To write

- One table: every symbol the course uses, in the spelling of the macros in `la_preamble.tex`.
- The collisions worth naming in print, all of them inherited from the linear-model notation the applied courses share: $\boldsymbol{\varepsilon}$ is the error vector and $\mathbf{e}_j$ the standard basis vector; $\boldsymbol{\beta}$ is the regression coefficient vector and $\mathbf{b}$ the right-hand side of a linear system; $\mathbf{D}_n$ is the duplication matrix and $\mathbf{D}_s$ the diagonal matrix of standard deviations.
- The Abadir and Magnus spellings that differ from this course's, so the textbook can be read alongside the notes without translation errors.